

How to predict the future with AI, one word at a time

CS50 Seminar, 2025

Sein Yun and Mohamed Zidan Cassim

What is AI?

What is AI?

- Does AI think?

What is AI?

- Does AI think? **No**

What is AI?

- Does AI think? **No**
- Is AI accurate?

What is AI?

- Does AI think? **No**
- Is AI accurate? **Sometimes**

What is AI?

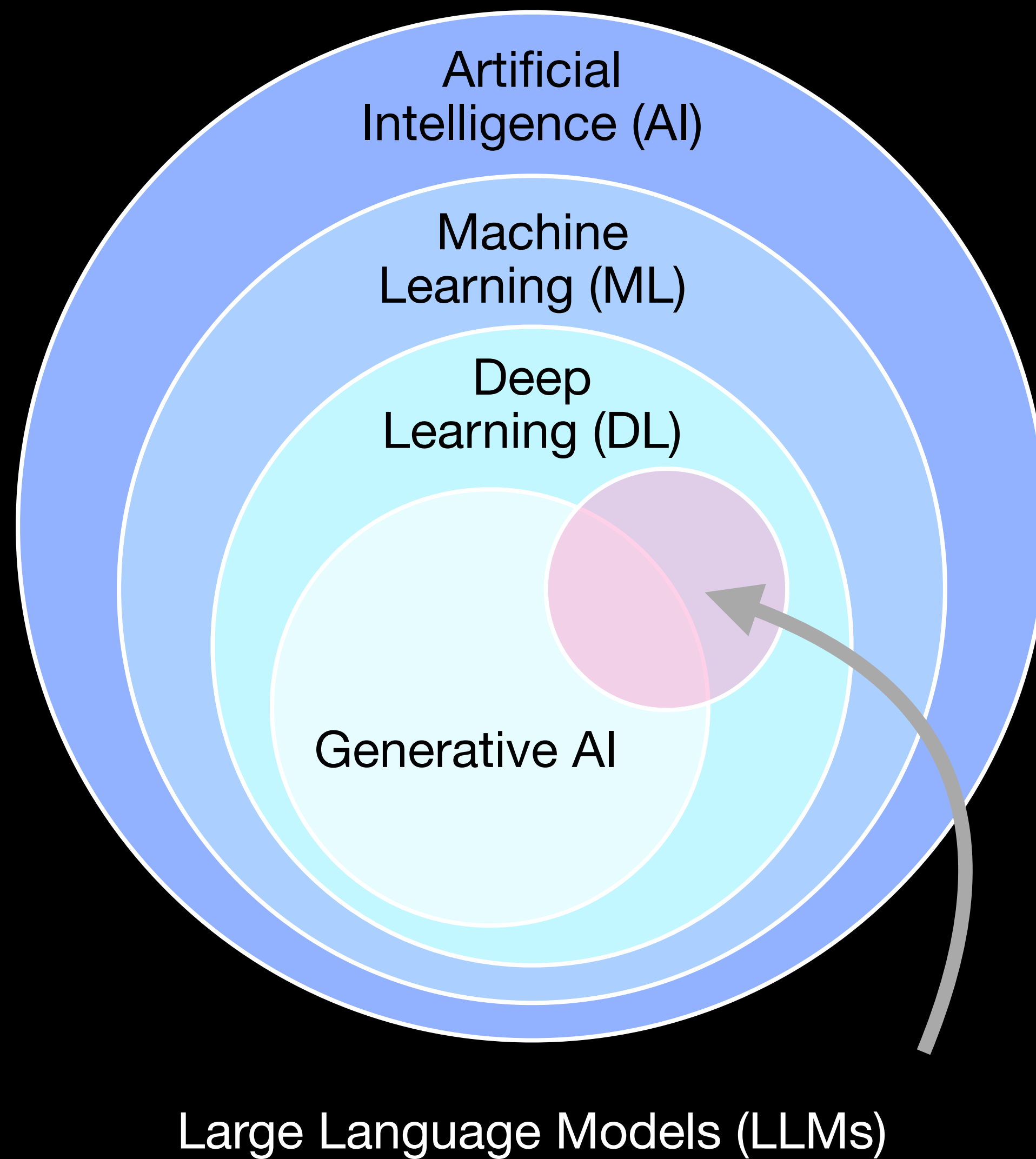
- Does AI think? **No**
- Is AI accurate? **Sometimes**

Is AI useful?

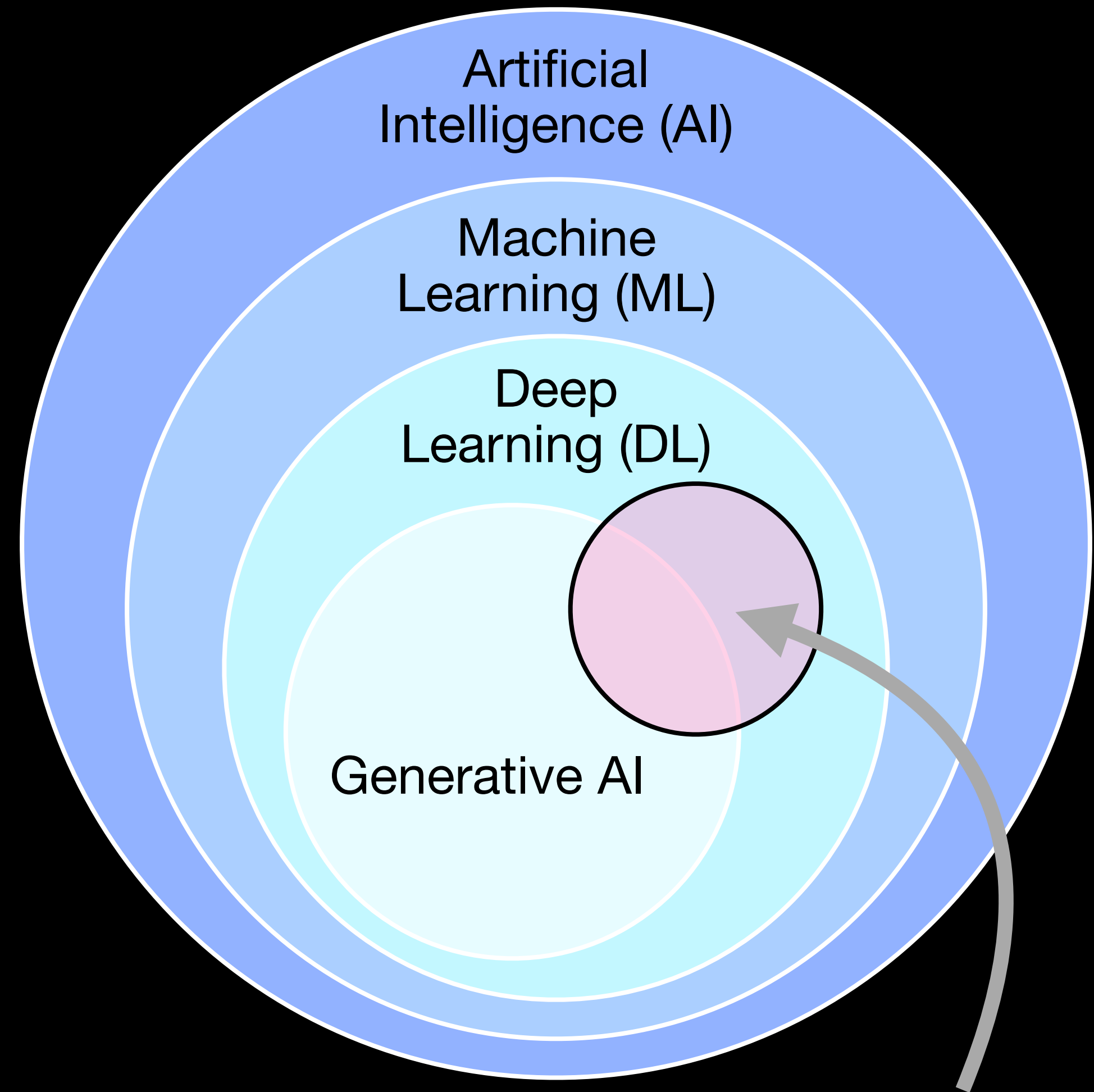
What is AI?

- Does AI think? **No**
- Is AI accurate? **Sometimes**
- Is AI useful? **Incredibly**

What is AI?



What is AI?



Large Language Models (LLMs)

How LLMs Work

How LLMs Work

Twinkle, twinkle, little ...

How LLMs Work

Twinkle, twinkle, little ~~duck?~~

How LLMs Work

Twinkle, twinkle, little ~~duck?~~ star.

How LLMs Work

Twinkle, twinkle, little star.

How LLMs Work

night

sun

plum

Twinkle, twinkle, little star.

duck

car

moon

How LLMs Work

Twinkle, twinkle, little star.

night

sun

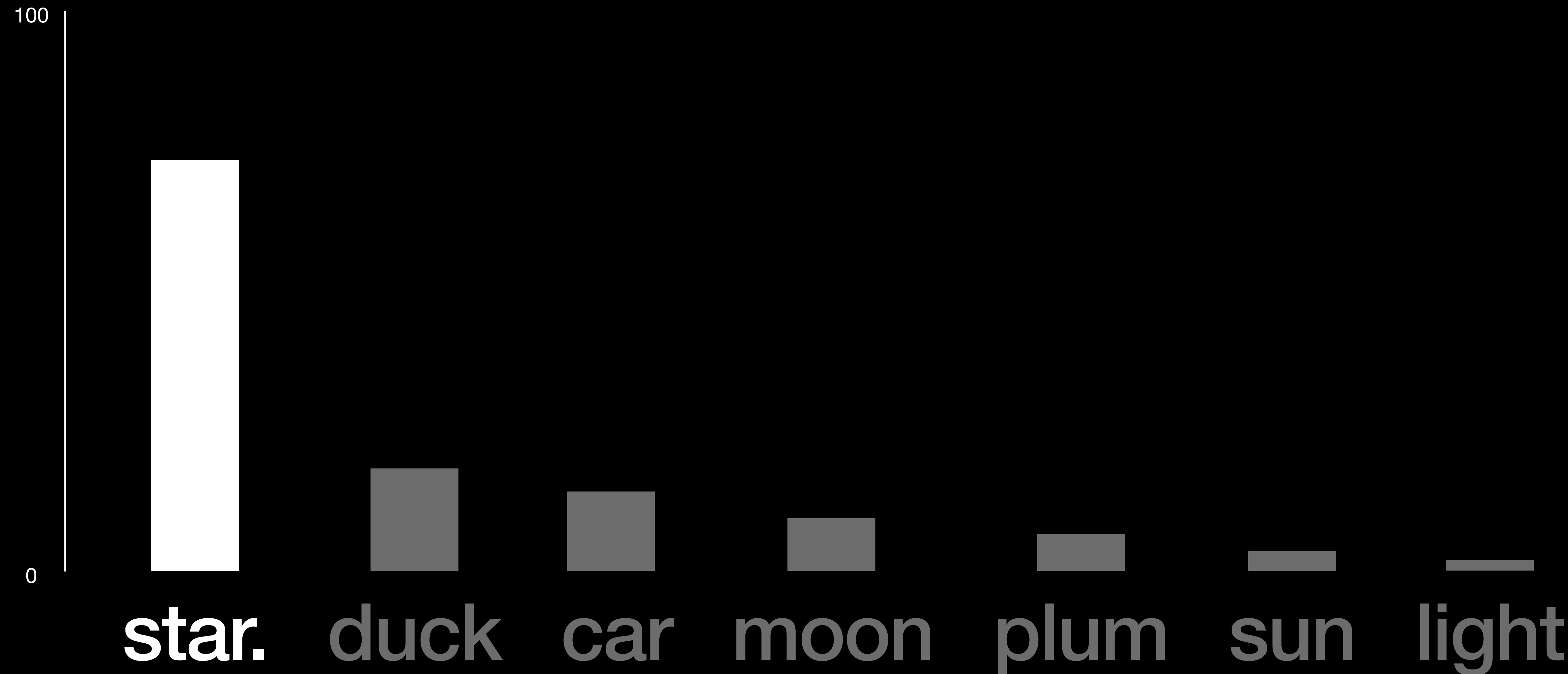
plum

duck

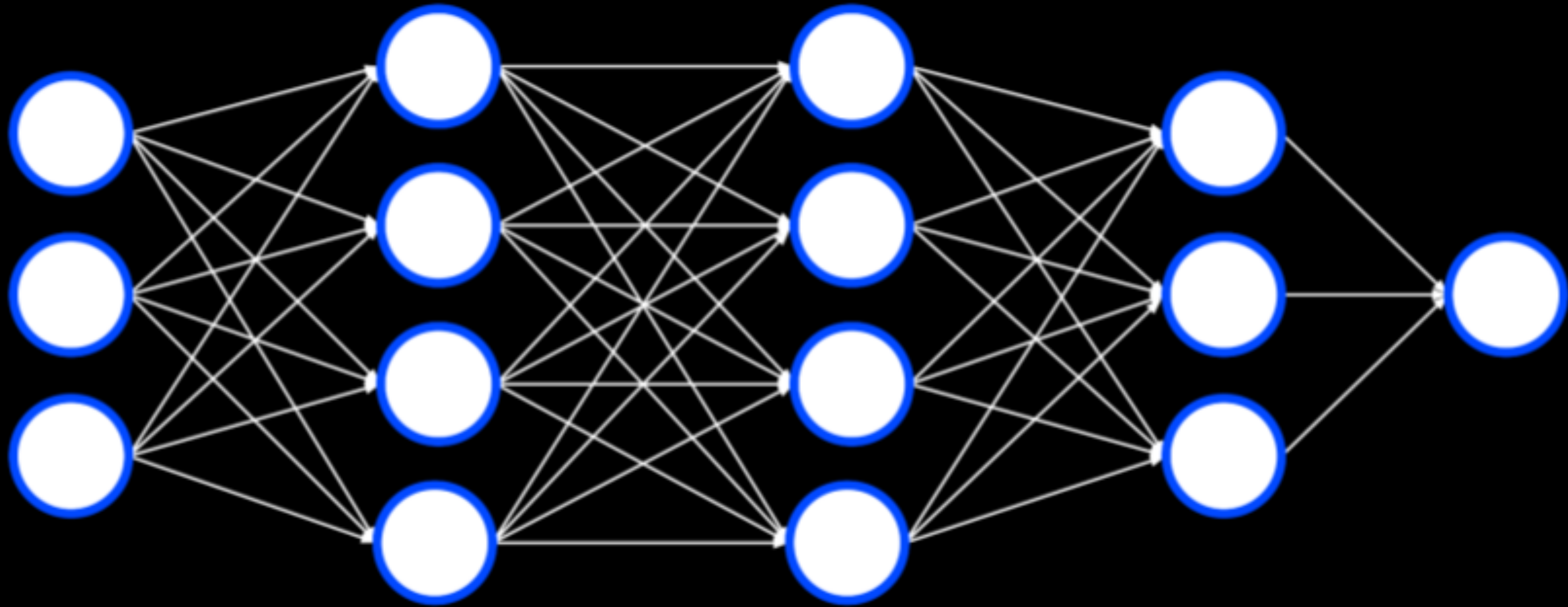
car

moon

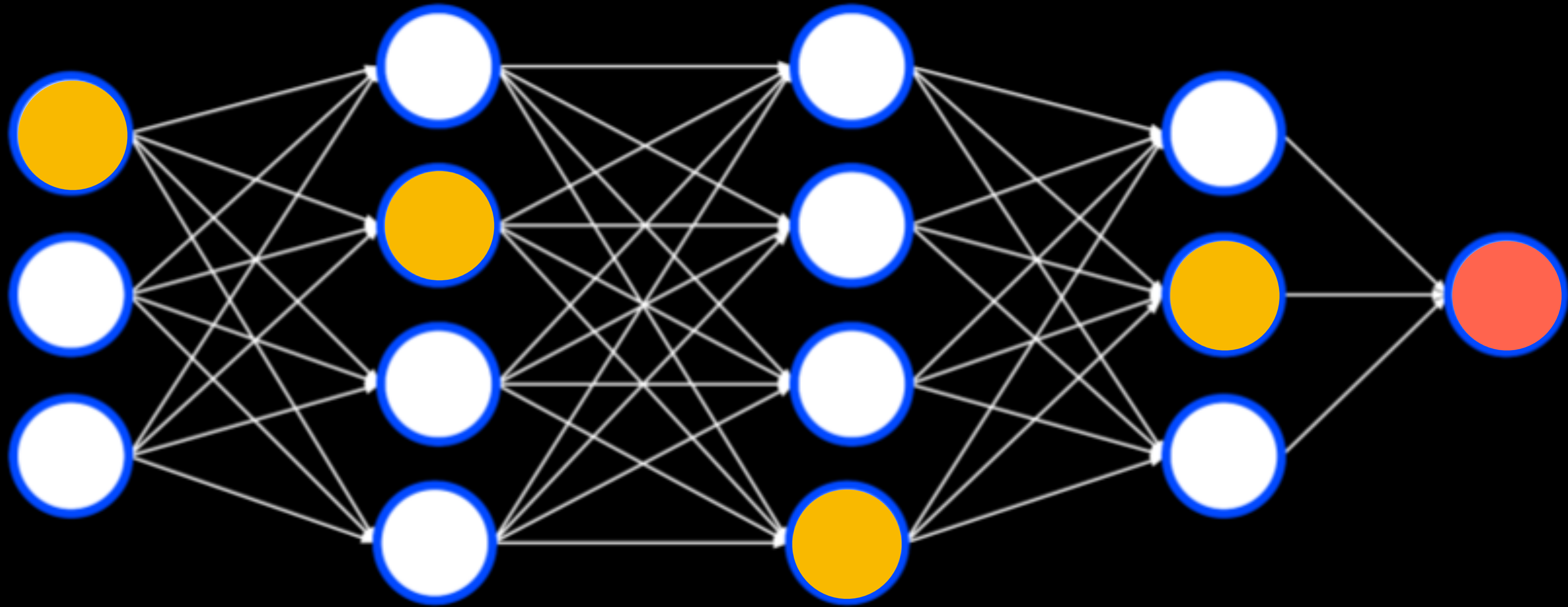
How LLMs Work



How LLMs Work

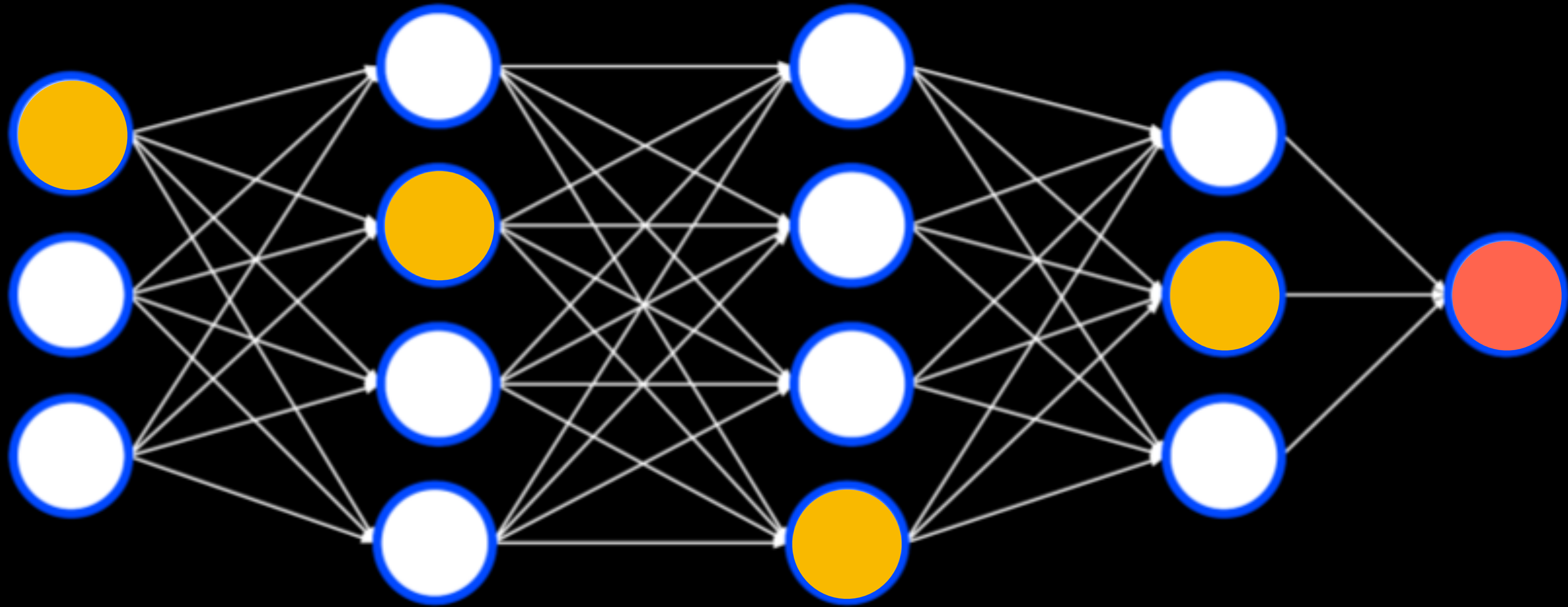


How LLMs Work



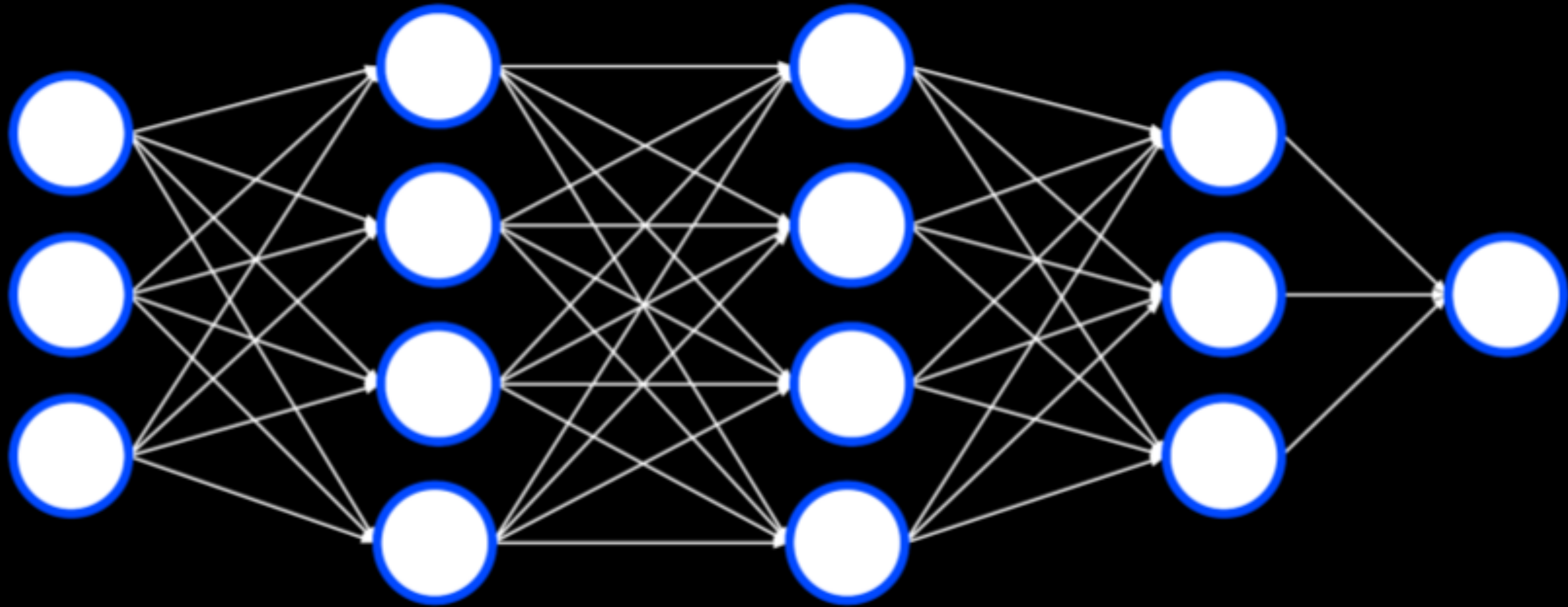
- **Fine Tuning:** Using high quality data to adjust the parameters to be more useful.

How LLMs Work



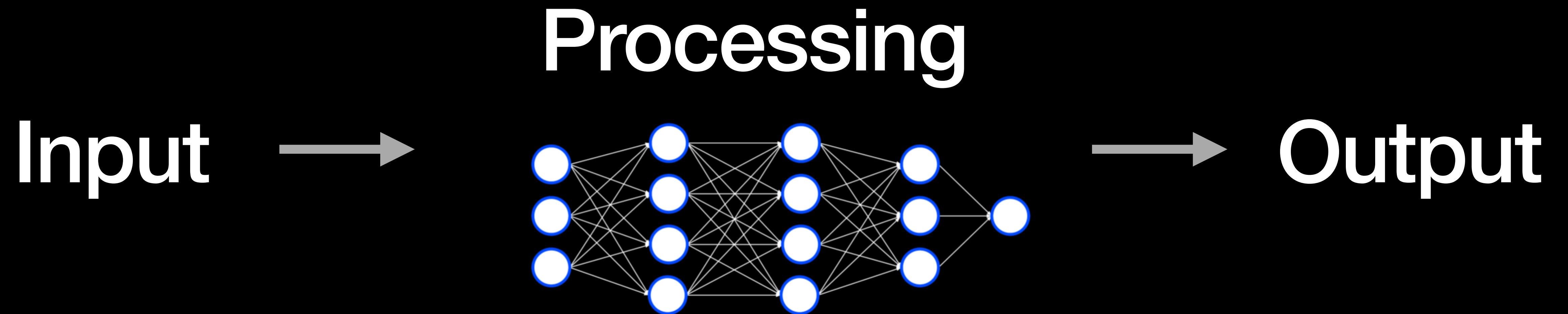
- **Fine Tuning:** Using high quality data to adjust the parameters to be more useful.

How LLMs Work



- **Fine Tuning:** Using high quality data to adjust the parameters to be more useful.

How LLMs Work



Limitations

Limitations

- Hallucinations
- Biased
- Susceptible to attacks

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- Biased
- Susceptible to attacks

Limitations

Hallucinations

To upgrade your RAM, remove it from the computer and soak it in warm water for 20 minutes to refresh the memory cells.

Limitations

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Limitations

Hallucinations

To upgrade your RAM, remove it from the computer and soak it in warm water for 20 minutes to refresh the memory cells.

Chain of Thought & Tooling

Chain of Thought

Standard input-
output prompting

Input → Output

Chain of Thought

Standard input-
output prompting

Input → Output

Chain of thought
prompting

Input → thought → ... → thought → Output

→ “Showing your work”

→ Improvements in decreasing hallucinations

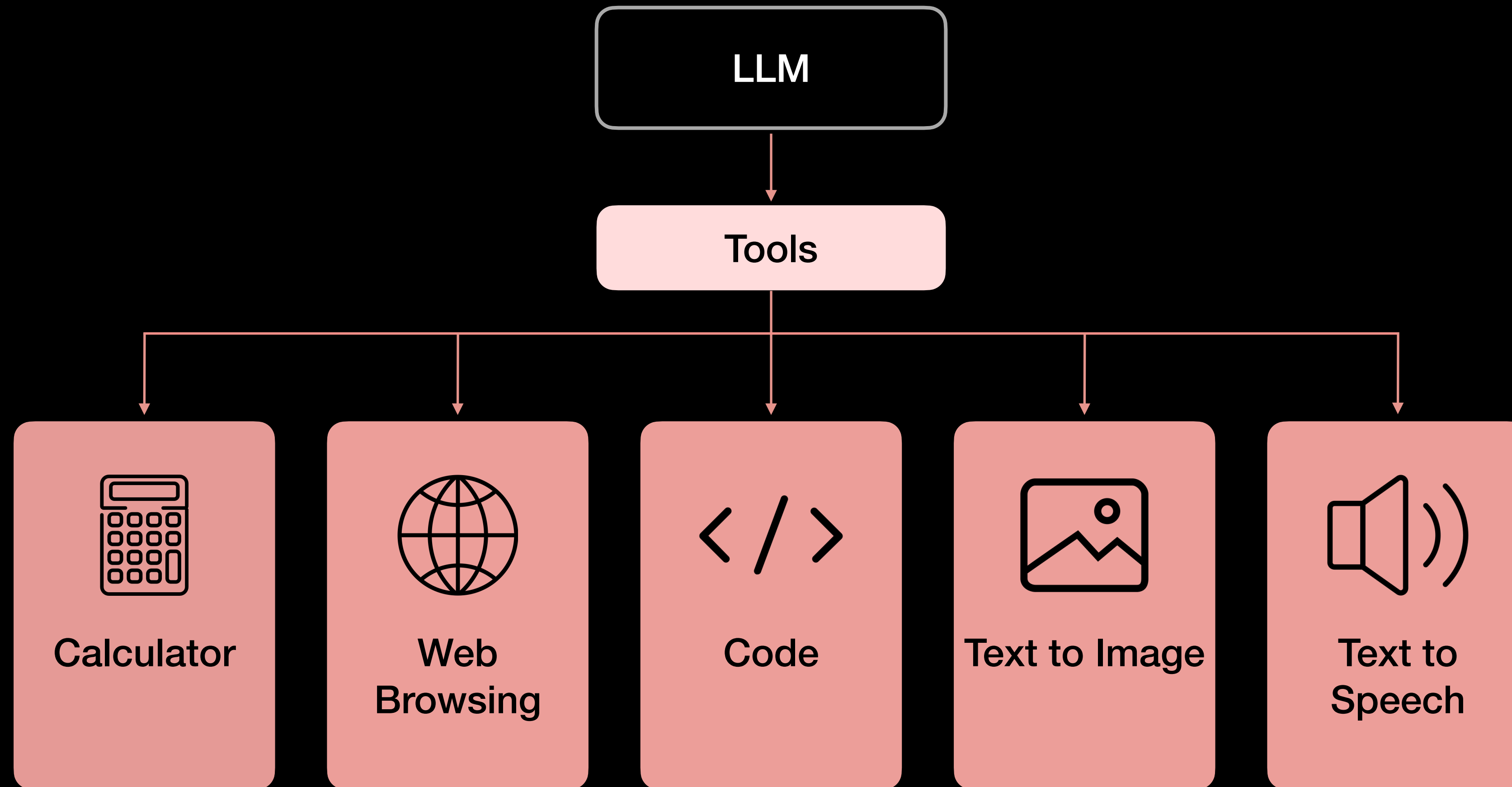
Tooling

Tooling ... because LLMs were
bad at math?!

$$2 + 2 =$$

$$2 + 2 = 5?$$

Tooling





Cursor AI



Windsurf AI

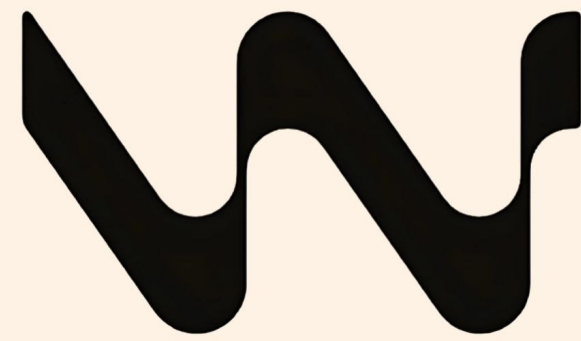
**CLAUDE
CODE**



Lovable

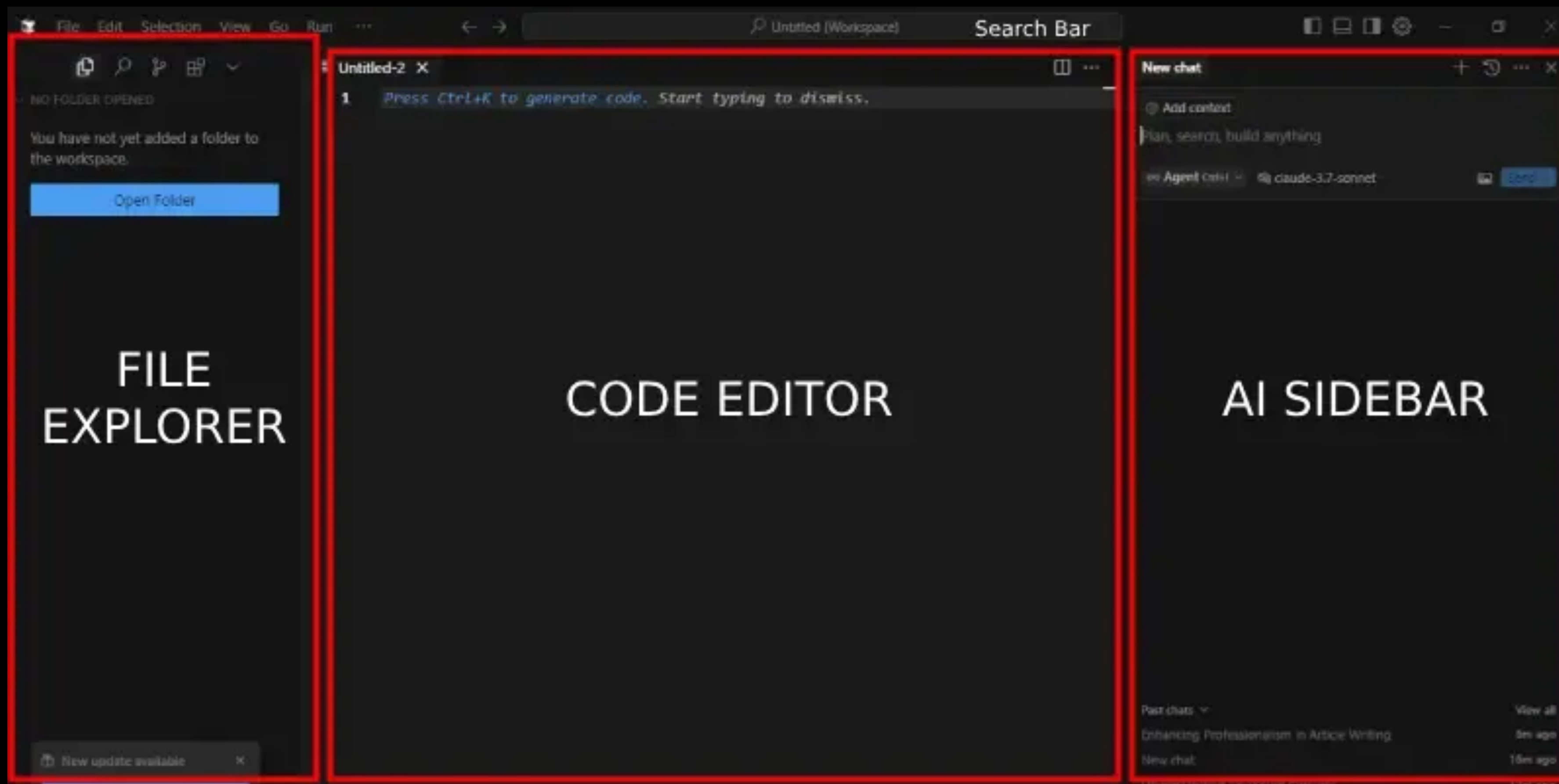


Cursor AI



Windsurf AI

CLAUDE
CODE



So how do I use AI?

1. Experiment

1. Experiment

2. Experiment

1. Experiment

2. Experiment

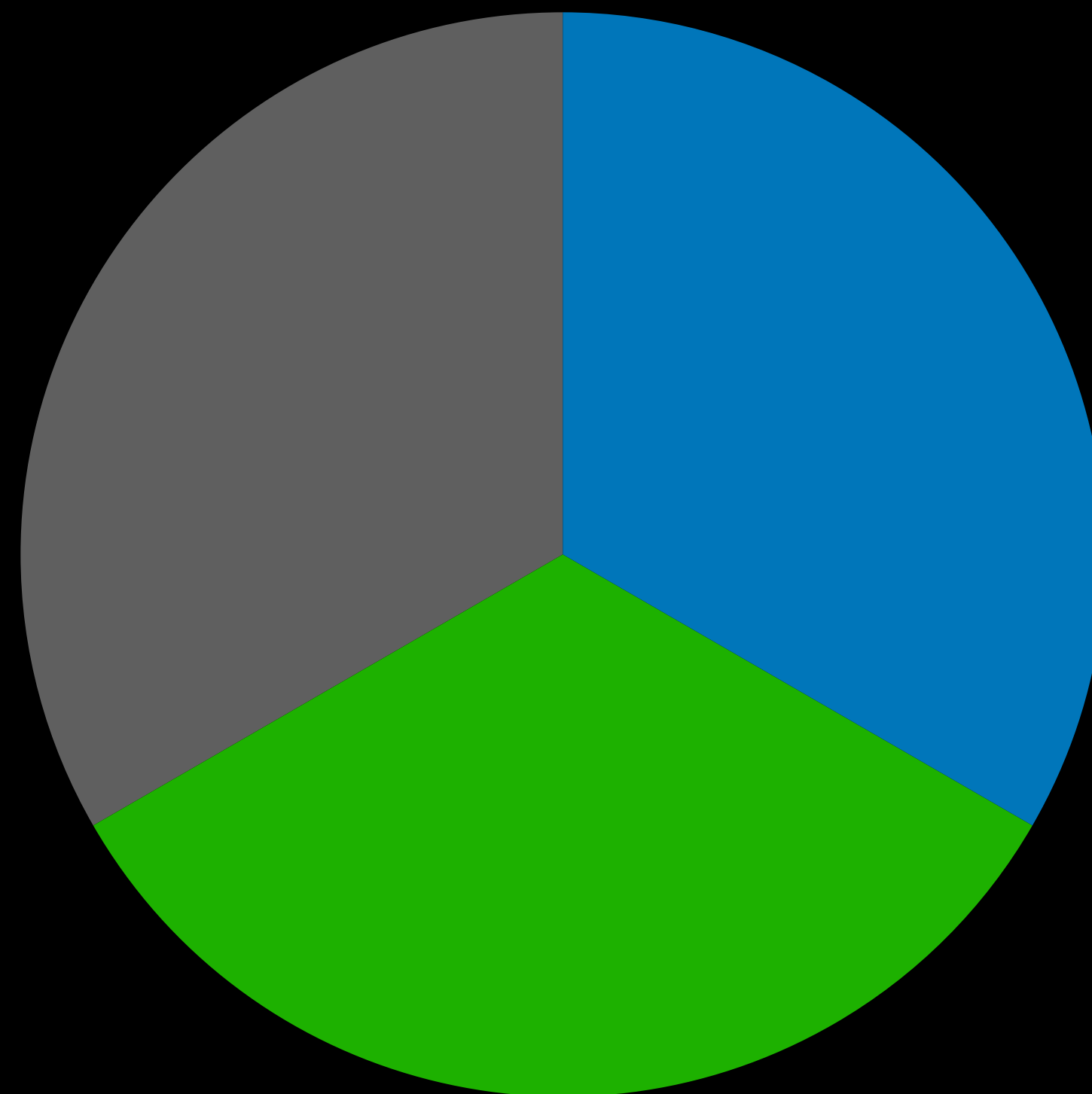
3. Experiment

1. Experiment

2. Experiment

3. Experiment

How do I use this AI thing?



- Experiment
- Experiment but in blue
- Experiment but in green

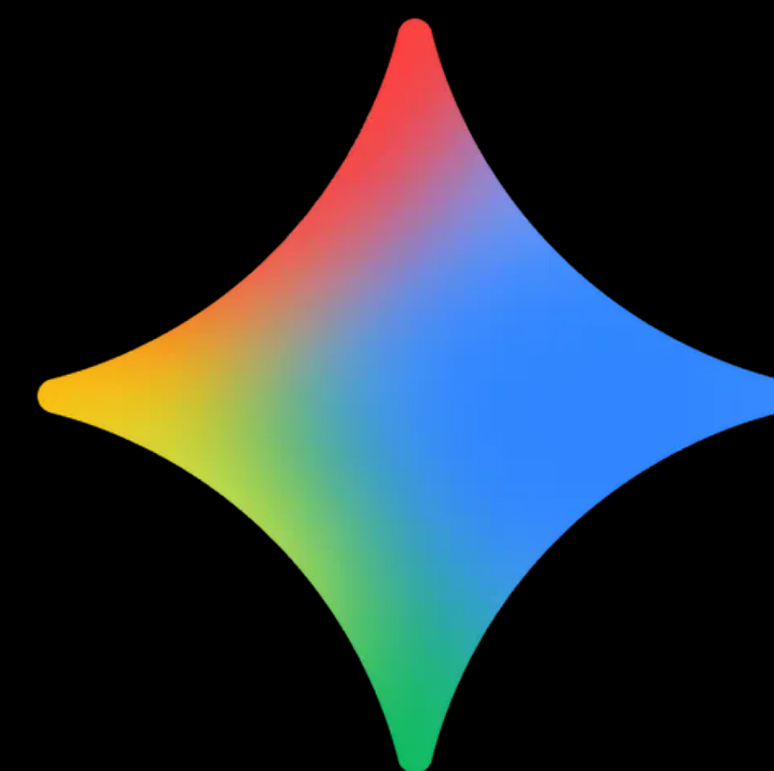
Which model should I use?



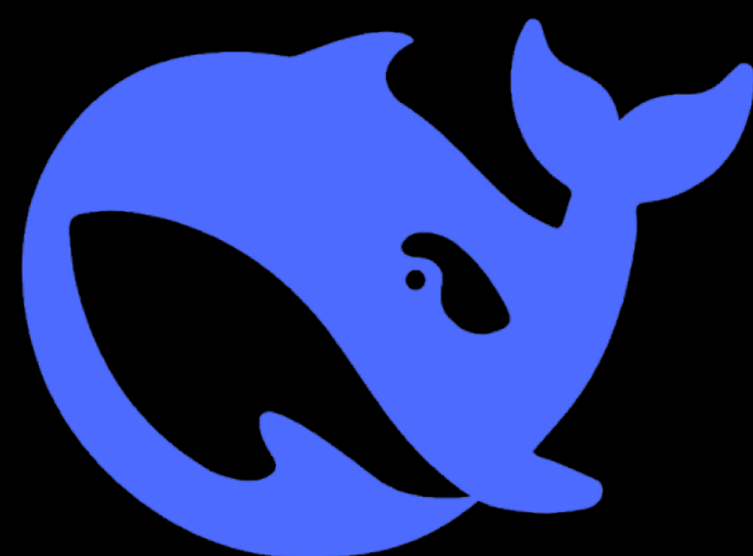
ChatGPT



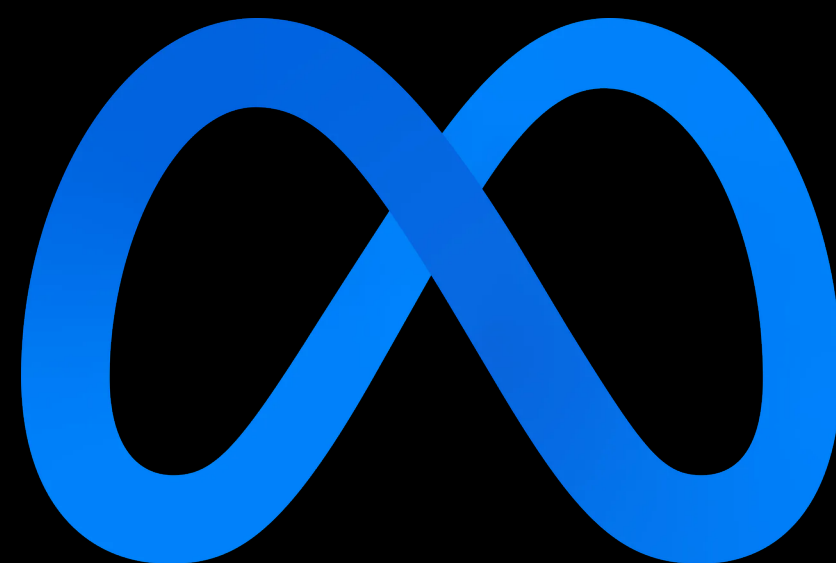
Claude



Google Gemini



DeepSeek



LLaMa



Grok

Overview

Text

WebDev

Vision

Text-to-Image

Image Edit

Search

Text-to-Video

Image-to-Video

Copilot

Start Voting

Leaderboard Overview

See how leading models stack up across text, image, vision, and beyond. This page gives you a snapshot of each Arena, you can explore deeper insights in their dedicated tabs. Learn more about it [here](#).

📄 Text

🕒 1 day ago

Rank (UB) ↑↓	Model ↑↓	Score ↓	Votes ↑↓
1	 gemini-2.5-pro	1452	61,259
1	 claude-opus-4-1-20250805-thi...	1448	27,970
1	 claude-sonnet-4-5-20250929-t...	1448	12,313
2	 gpt-4.5-preview-2025-02-27	1442	14,644
3	 claude-opus-4-1-20250805	1439	40,413
3	 chatgpt-4o-latest-20250326	1438	46,917
4	 gpt-5-high	1436	29,552
2	 claude-sonnet-4-5-20250929	1436	5,483
4	 o3-2025-04-16	1435	57,953
4	 qwen3-max-preview	1433	24,457
View all			

📄 WebDev

🕒 1 day ago

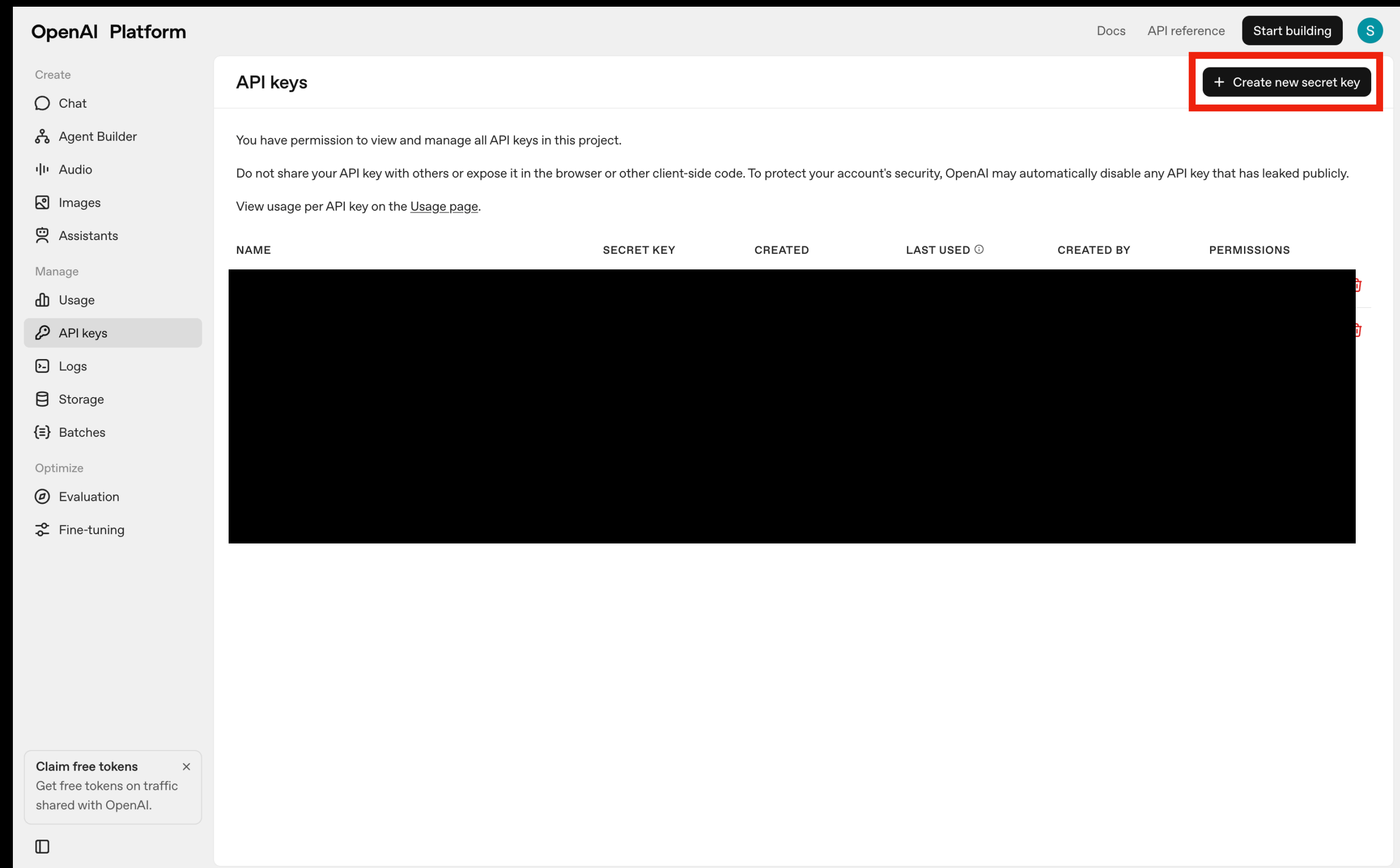
Rank (UB) ↑↓	Model ↑↓	Score ↓	Votes ↑↓
1	 GPT-5 (high)	1473	8,004
1	 Claude Opus 4.1 thinking-16k...	1458	8,726
2	 Claude Opus 4.1 (20250805)	1451	8,986
4	 Claude Sonnet 4.5 (thinking ...	1420	4,863
4	 MiniMax-M2	1405	3,515
5	 Gemini-2.5-Pro	1399	14,628
5	 GLM-4.6	1395	7,563
5	 DeepSeek-R1-0528	1393	4,800
6	 Claude Sonnet 4.5	1387	7,855
7	 Claude Opus 4 (20250514)	1383	9,238
View all			

Application: API Keys

How to generate an OpenAI API key

How to generate an OpenAI API key

- <https://platform.openai.com/api-keys>



Create new secret key

Owned by

You

Service account

This API key is tied to your user and can make requests against the selected project. If you are removed from the organization or project, this key will be disabled.

Name Optional

My Test Key

Project

Default project

Permissions

All

Restricted

Read only

Cancel

Create secret key

Create new secret key

Owned by

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Service account

This API key is tied to your user and can make requests against the selected project. If you are removed from the organization or project, this key will be disabled.

Name Optional

My Test Key

Project

Default project



Permissions

All

Restricted

Read only

Cancel

Create secret key

****Save your API key somewhere!**



final-project

test_samples.csv

feature_extractor.py

.env

evaluate_model.py

common_misclassifications.csv

EXPLORER

FINAL-PROJECT

src

data_loader.py

evaluate_model.py

extract_features.py

feature_extractor.py

plot_per_class_metrics.py

train_classifier.py

train_models.py

.env

.gitattributes

.gitignore

analyze_samples.py

documentation.md

generate_samples.py

README.md

requirements.txt

OUTLINE

No symbols found in document '.env'

TIMELINE .env

Local History will track recent changes as you save them unless the file has been excluded or is too large.

.env

1 OPENAI_API_KEY=

2 ANTHROPIC_API_KEY=

3 LLAMA_API_KEY=

4 LLAMA_API_ENDPOINT=

main*

Launchpad

0 0

Live Share

-- NORMAL --

Ln 4, Col 42

Spaces: 4

UTF-8

LF

{ } Properties

Go Live

Prettier

Pricing / Billing

M Mohamed / tldr

DashboardDocsAPI reference

M

Settings

Your profile

Organization

General

API keys

Admin keys

People

Projects

Billing

Limits

Usage

Service health

Data controls

Security

Project

General

API keys

Webhooks

Evaluations

People

</> Cookbook

Forum

Billing

OverviewPayment methodsBilling historyCredit grantsPreferences

Pay as you go

Credit balance

Add to credit balanceCancel plan

Your credit balance will be consumed as you use the API. Visit the [usage page](#) to view a breakdown of your consumption.

Auto recharge is off

When your credit balance reaches \$0, your API requests will stop working. Enable automatic recharge to automatically keep your credit balance topped up.

Enable auto recharge

Payment methods

Add or change payment method

Billing history

View past and current invoices

Preferences

Manage billing information

Usage limits

Set monthly spend limits

Pricing

View pricing and FAQs

Flagship models

Our frontier models are designed to spend more time thinking before producing a response, making them ideal for complex, multi-step problems.

GPT-5

The best model for coding and agentic tasks across industries

Price

Input:
\$1.250 / 1M tokens

Cached input:
\$0.125 / 1M tokens

Output:
\$10.000 / 1M tokens

GPT-5 mini

A faster, cheaper version of GPT-5 for well-defined tasks

Price

Input:
\$0.250 / 1M tokens

Cached input:
\$0.025 / 1M tokens

Output:
\$2.000 / 1M tokens

GPT-5 nano

The fastest, cheapest version of GPT-5—great for summarization and classification tasks

Price

Input:
\$0.050 / 1M tokens

Cached input:
\$0.005 / 1M tokens

Output:
\$0.400 / 1M tokens

GPT-5 pro

The smartest and most precise model

Price

Input:
\$15.00 / 1M tokens

Cached input:
-

Output:
\$120.00 / 1M tokens

Demo

